

# **Learned exchange-correlation functionals**

**Training-subset generation, pre-training, in-sample optimization and held-out generalization**

2026-09-15

# The objects named in this talk

- **Cell:** one trained exchange-correlation functional (an exchange and a correlation network), identified by its architecture and the number  $r$  of training points it was fitted to. **Subset  $r$ :** the  $r$  points of the 26-point DFS pool [DFS21] chosen for that column of cells (Sec. 2).
- **Run:** one cluster run under one configuration file; the runs differ in which architectures they train. **View:** the runs merged into one figure set. **Arm:** a run of one architecture (deep\_3x16) under a changed training protocol, shown with its tag: [25 cycles], [dpyscf parity].
- **Pre-training:** fitting a network to reproduce its parent functional (PBE or SCAN) before any reference data is seen. **Certificate:** the check that the clone reproduces the parent to tolerance; a failing certificate holds the group's training.
- **Checkpoints:** training saves the final, the lowest-training-loss and the lowest-validation-error network; every figure reads the **validation-best** one.
- **Rung:** the class of density variables a functional reads (GGA: density and gradient; meta-GGA: plus the kinetic-energy density; rung 3.5: plus density-matrix columns). **Start:** deep networks begin from a Glorot-initialized last layer, deep0 networks from a zeroed one, so the untrained deep0 network is the LDA.

## The measures named in this talk

- **WTMAD-2:** GMTKN55's weighted total mean absolute deviation of reaction energies [Goe17]: each benchmark's mean absolute error divided by its mean absolute reference value and scaled by 56.84 kcal/mol; two benchmarks are present here (BH76 barriers, W4-11 atomization energies). Lower is better; the pooled PBE value is 8.92 kcal/mol.
- $\epsilon_n$ : the DFS density error [DFS21 Eq. 20], the integrated absolute difference from the CCSD reference density per electron (PBE 0.00923). **Grid RMSE:** the root-mean-square density difference over the quadrature grid, the measure the density loss trains on.
- $\mathcal{ED}$ : the harmonic mean of the energy error and  $\gamma$  times the density error [DFS21 Eq. 21]; **DFS units** use  $\epsilon_n$  with  $\gamma = 1084.87$  kcal/mol, **self-calibrated units** the grid RMSE with  $\gamma$  set so that PBE's two legs are equal.
- **Beats PBE / cap:** a cell is below PBE when its  $\mathcal{ED}$  is below PBE's on the same held-out slice; on the bar charts PBE's value on the cell's own slice is the black cap and a triangle marks a cell below it.
- **Recurring tail:** the species whose density error exceeds 1.5 times PBE's in at least half of the cells carrying them and in at least two (eight species). **T1:** the CCSD T1 diagnostic of multireference character; 38 of the 189 held-out species are above 0.02, the model-free counterpart of the tail.

# Scope

One network pair (exchange and correlation) per architecture and training-subset size, trained in separate cluster runs under one protocol and drawn once over every finished cell. Each run carries a pre-training, the certificate that gates its training array, and the array itself, one cell per (architecture, subset size).

| Run               | Architectures                                      | Cells in the figures               |
|-------------------|----------------------------------------------------|------------------------------------|
| size              | deep_3x16, deep_attn_3x16, deep_2x8, deep_attn_2x8 | 41 of 44                           |
| families          | deep0_3x16, deep0_attn_3x16, deep0_cusp_3x16       | 19 of 33                           |
| meta-GGA families | five SCAN-parented architectures                   | 0 of 55 (four FAIL, one cancelled) |
| the two arms      | deep_3x16 [25 cycles], deep_3x16 [dpyscf parity]   | 2 of 5 each                        |

- Subset sizes 1, 2, 3, 4, 5, 6, 7, 12, 15, 18, 26 points of the 26-point DFS pool [DFS21, SI Sec. II], chosen by Jensen-Shannon selection (Sec. 2).
- 64 cells in the figures; two evaluated cells (deep\_attn\_2x8 at 18, deep\_3x16 [25 cycles] at 15) are set aside pending re-evaluation (Sec. 6).
- Basis 6-311++G(3df,2pd), grid level 3, density fitting, at every stage [cfg]; the validation-best checkpoint is read for every optimization figure.

## 1. Model: density variables and the DFS coordinates

With  $n = n_\alpha + n_\beta$  and  $\sigma = |\nabla n|^2$  [DFS21 Eqs. 4, 5]:

$$k_F = (3\pi^2 n)^{1/3}, \quad s = \frac{|\nabla n|}{2k_F n}, \quad r_s = \left(\frac{3}{4\pi n}\right)^{1/3},$$

$$\zeta = \frac{n_\alpha - n_\beta}{n}, \quad \phi(\zeta) = \frac{1}{2} \left[ (1 + \zeta)^{4/3} + (1 - \zeta)^{4/3} \right].$$

The networks read logarithmic compressions of these [DFS21 Eqs. 7 to 10; code: networks.py:621-657]:

$$x_0 = \ln(n^{1/3} + 10^{-5}), \quad x_1 = \ln \phi(\zeta), \quad x_s = (1 - e^{-s^2}) \ln(s + 1), \quad x_\alpha = \ln \frac{\alpha + 1}{2}.$$

- Exchange input  $[x_s, \dots]$  with  $s$  of the doubled spin channel  $2n_\sigma$  (the exchange network is evaluated per spin channel, next slides); correlation input  $[x_0, x_1, x_s, \dots]$  on the total density. The ellipsis holds the architecture's extra descriptors; on the meta-GGA rung the raw  $\alpha$  is replaced by  $x_\alpha$ .
- The  $10^{-5}$  in  $x_0$  is the DFS reference implementation's constant [dpyscf net.py:39; code: networks.py:27].

## 1. Model: the extra descriptors

**Iso-orbital indicator** (meta-GGA rung; SCAN's  $\alpha$  [SCAN15 Eq. 2; DFS21 Eq. 6; code: `metagga.py:8-9`]), a contraction of the live density matrix  $P$  against the AO gradients, hence self-consistent and differentiable through the SCF:

$$\alpha = \frac{\tau - \tau_W}{\tau_{\text{unif}}}, \quad \tau = \frac{1}{2} \sum_i |\nabla \psi_i|^2 = \frac{1}{2} \sum_{\mu\nu} P_{\mu\nu} \nabla \chi_\mu \cdot \nabla \chi_\nu,$$

$$\tau_W = \frac{|\nabla n|^2}{8n}, \quad \tau_{\text{unif}} = \frac{3}{10} (3\pi^2)^{2/3} n^{5/3},$$

with  $\alpha \geq 0$  imposed by a smooth positive part rather than a hard clip [code: `metagga.py:25-31`].

**Nuclear cusp pair** (`cusp`), two bounded geometric features per grid point [code: `descriptors.py:230-245`]:

$$x_4 = \exp(-2Z_{\text{nearest}} r_{\min}), \quad x_5 = \tanh\left(\frac{1}{5} \ln \sum_A \frac{Z_A}{r_A}\right),$$

$x_4$  the density-form cusp envelope  $e^{-2Zr}$  [Steiner63, after Kato57],  $x_5$  the log-compressed bare-nuclear potential. Both are geometric, so the synthetic pre-training mesh (Sec. 3) carries no values for them.

# 1. Model: the network

Each of the exchange and the correlation networks is a multilayer perceptron of depth hidden layers of nodes units with GELU activations, read at every grid point; its scalar output  $g$  becomes the enhancement factor on the next slide [code: networks.py:169, 379–404].

- **GELU:**  $\text{GELU}(z) = z\Phi(z)$  with  $\Phi$  the standard normal distribution function, a smooth rectifier; the enhancement factors and their derivatives (the potential) are smooth in the density variables.
- **Last layer:** Glorot-initialized in the deep networks, zeroed in the deep0 networks, so the latter start exactly at  $F = 1$  (the LDA) before pre-training.
- **Attention variants** (`_attn`) [Vas17 Eq. 1; Xio20; code: net.py:55–136]: after the first hidden layer, the  $H$  hidden units are  $h$  tokens of dimension  $d_k = H/h$  under pre-layer-norm self-attention across the token axis, a per-point re-mixing of hidden units; the functional stays semilocal.

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V, \quad x \leftarrow x + W^O \text{Concat}(\text{head}_1, \dots, \text{head}_h)(\text{LN}(x)).$$

# 1. Model: the functional form

MLP output  $g$  at a grid point:

$$F_x = 1 + \mathcal{L}_{1.804}(G_x \cdot g_x), \quad F_c = 1 + \mathcal{L}_2(G_c \cdot g_c),$$

$$\mathcal{L}_\lambda(z) = \lambda \operatorname{sigmoid}(z - \ln(\lambda - 1)) - 1 \quad [\text{DFS21 Eq. 11}],$$

$$G = \tanh^2(s) \text{ (GGA)}, \quad G = x_s + \tanh^2(x_\alpha) \text{ (meta-GGA) [DFS21 Eq. 12].}$$

$\mathcal{L}_\lambda(0) = 0$  and  $F \in (0, \lambda)$ ;  $\lambda = 1.804 = 1 + \kappa_{\text{PBE}}$  for exchange [PBE96 Eq. 14] (DFS uses 1.174 [PRSB14]),  $\lambda = 2$  so that  $F_c \geq 0$  [DFS21 Eq. 13]. The gate  $G$  pins  $F = 1$  where the gradient vanishes.

$$E_{xc} = \frac{1}{2} (E_x[2n_\alpha] + E_x[2n_\beta]) + E_c[n_\alpha, n_\beta] \quad [\text{OP79; code: descriptors.py:124}],$$

$$E_x[n] = \int n \epsilon_x^{\text{LDA}}(n) F_x d^3r, \quad E_c = \int n \epsilon_c^{\text{PW92}}(r_s, \zeta) F_c d^3r \quad [\text{PW92}].$$

# 1. Model: the architectures

| Architecture    | Last layer | Rung, parent | Hidden | Heads | Extra |
|-----------------|------------|--------------|--------|-------|-------|
| deep_3x16       | Glorot     | GGA, PBE     | 3 x 16 | none  | none  |
| deep_attn_3x16  | Glorot     | GGA, PBE     | 3 x 16 | 4     | none  |
| deep0_3x16      | zeroed     | GGA, PBE     | 3 x 16 | none  | none  |
| deep0_attn_3x16 | zeroed     | GGA, PBE     | 3 x 16 | 4     | none  |

- Inputs:  $F_x$  from  $x_s$ ,  $F_c$  from  $x_0, x_1, x_s$ ; the cusp architectures add  $x_4, x_5$  to both, the meta-GGA ones  $x_\alpha$  as well, the rung-3.5 ones their occupancy columns.
- The 2x8 pair, deep\_2x8 and deep\_attn\_2x8 (2 heads), are the same GGA networks at 2 x 8; the arms' cells are deep\_3x16 trained under the 25-cycle and the dpyscf-parity protocol, shown on the same axis as deep\_3x16 [25 cycles] and deep\_3x16 [dpyscf parity].
- deep\_3x16 and deep0\_3x16 have the same layers, inputs and parameter count and differ only in the start; after pre-training both are PBE clones, so the two columns are replicates of one architecture under two starts: their combined held-out verdicts agree at every subset size (Sec. 5).

## 2. Subsets: the 26-point DFS pool [DFS21 SI Sec. II]

- 21 atomization energies (G2/97 molecules), 3 BH76 reaction energies and 2 IP13 ionization potentials: the training set of the Letter, 26 points over 30 unique species.
- Each point carries the species its loss term needs: an atomization its molecule and the H and Li anchors it contains, a reaction every reactant and product, an ionization the neutral and the cation.
- References: atomization energies [HK12] with H<sub>2</sub>O and C<sub>2</sub>H<sub>2</sub> from [W411]; reaction energies [Goe17, BH76RC]; ionization potentials [NIST]. Code: `dfs_pool.py`, `training_points.py`:347-420.
- The pool is indexed 0 to 25 in the selection ledger; the chosen subsets of the next slides are lists of these indices.

► the 26 points

## 2. Subsets: the descriptor distributions

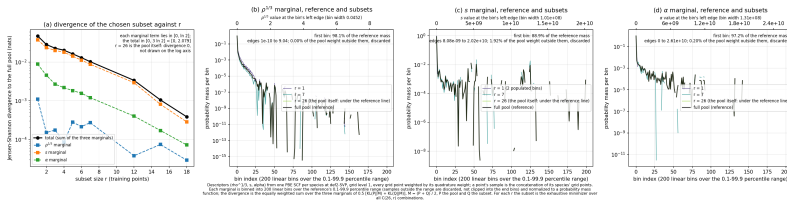
One PBE SCF per species (def2-SVP, grid level 1) [code: subset\_selection.py:106–118, 448–470] gives at every grid point  $g$  with quadrature weight  $w_g$

$$\rho^{1/3}, \quad s = \frac{|\nabla \rho|}{2(3\pi^2)^{1/3} \rho^{4/3}} \text{ [PBE96]}, \quad \alpha = \frac{\tau - \tau_W}{\tau_{\text{unif}}} \text{ [SCAN15], clipped to } [0, 100].$$

A point's sample is the concatenation of its species' grid points; the reference is the concatenation over all 26 points. Each marginal is binned into 200 equal-width bins between its 0.1 and 99.9 percentiles and normalized to a mass function:

$$h_x^{(b)} = \sum_{g: x_g \in \text{bin } b} w_g, \quad P_x^{(b)} = \frac{h_x^{(b)}}{\sum_{b'} h_x^{(b')}}.$$

Training-subset selection: jensen-Shannon divergence to the full DFS pool



## 2. Subsets: the metric [Lin91]

The distance between a candidate subset  $S$  and the pool is the Jensen-Shannon divergence [code: subset\_selection.py:213–246, 472] summed over the three marginals with equal weights,  $P_x$  the pool and  $Q_x$  the subset mass function, probabilities floored at  $10^{-12}$ , natural logarithm:

$$\text{JSD}(S) = \sum_{x \in \{\rho^{1/3}, s, \alpha\}} \frac{1}{2} [\text{KL}(P_x \parallel M_x) + \text{KL}(Q_x \parallel M_x)], \quad M_x = \frac{1}{2}(P_x + Q_x),$$

$$\text{KL}(P \parallel Q) = \sum_b P^{(b)} \ln \frac{P^{(b)}}{Q^{(b)}}, \quad \text{each marginal term in } [0, \ln 2].$$

- The linear edges reach the low-density tails ( $s$  to  $2.0 \times 10^{10}$ ,  $\alpha$  to  $2.6 \times 10^{10}$ ,  $\rho^{1/3}$  from  $10^{-10}$  to 9.04), so the first bin holds 98.1, 88.9 and 97.2 percent of the weighted reference mass: the whole chemical range for  $s$  and  $\alpha$ , the vacuum tail ( $\rho < 9.2 \times 10^{-5}$ ) for  $\rho^{1/3}$ .
- The metric therefore compares the low-density tails of the molecular grids ( $s$ ,  $\alpha$ ) and the valence and core region ( $\rho^{1/3}$ ); the  $s$  term is 74 to 89 percent of every divergence.

## 2. Subsets: the search

- For each  $r \in \{1, \dots, 7, 12, 15, 18, 26\}$  the subset is the exhaustive minimizer of  $\text{JSD}(S)$  over all  $\binom{26}{r}$  combinations: 26 candidates at  $r = 1$ , 9.7 million at  $r = 12$ , one at  $r = 26$  (the whole pool).
- Histograms add over disjoint samples, so every point's histogram is binned once and a candidate's mass function is a sum; a candidate with no grid point inside a marginal's range gets an infinite divergence and is never chosen.
- The subsets are nested only where the minimizer happens to nest: the 2-to-6-point subsets are barriers and ionizations with one or two molecules; from seven points on, atomization energies dominate (next slide).
- The L2 distance was computed and stored beside the JSD but no training run used it.

## 2. Subsets: the chosen subsets [ledger subset\_index\_log.json; plot\_subset\_jsd.py]

| <i>r</i> | Points (ledger index order)                                                                  | AE / BH76 / IP | JSD    | $\rho^{1/3}$ | <i>s</i> | $\alpha$ |
|----------|----------------------------------------------------------------------------------------------|----------------|--------|--------------|----------|----------|
| 1        | H2O                                                                                          | 1 / 0 / 0      | 0.0475 | 0.00109      | 0.03764  | 0.00881  |
| 7        | CHN, CO2, HLi, CH, OH+N2_to_H+N2O, Li_IP, C_IP                                               | 4 / 1 / 2      | 0.0102 | 0.00027      | 0.00877  | 0.00120  |
| 12       | CO2, C2H2, CO, HLi, CH, NO2, CH2, OH+N2_to_H+N2O, OH+CH3_to_O+CH4, HF+F_to_H+F2, Li_IP, C_IP | 7 / 3 / 2      | 0.0034 | 0.00004      | 0.00292  | 0.00040  |
| 26       | the whole pool                                                                               | 21 / 3 / 2     | 0      | 0            | 0        | 0        |

- The  $\rho^{1/3}$ , *s* and  $\alpha$  columns are the three marginal terms of the JSD (their sum is the total); every value is recomputed from the descriptor caches and agrees with the ledger to  $10^{-9}$ .
- The 3-to-6-point subsets hold the two barriers and the Li ionization with at most one or two molecules (the 2-point one is LiH with one barrier); the cells trained on 2 to 6 points generalize worst (Sec. 5). The subsets at 15 and 18 add molecules to the twelve-point set and do not nest between 6 and 7.

► all eleven subsets

### 3. Pre-training: targets and objective

Each network is fitted pointwise [code: pretrain.py:261-341, pretrain\_data\_gen.py] to reproduce its parent (PBE on the GGA rung, SCAN on the meta-GGA) on the parent's own densities of the DFS pre-training inventory plus every atom of the BH76 and W4-11 pools (38 systems on the GGA rung, 36 on the meta-GGA; the certificate re-evaluates 39 and 38), with a synthetic  $(r_s, s, \alpha)$  mesh [DFS21 SI] at 30 percent of the weight:

$$t_x = F_x^{\text{parent}} - 1 = \frac{\epsilon_x^{\text{parent}}}{\epsilon_x^{\text{LDA}}} - 1, \quad t_c = F_c^{\text{parent}} - 1 = \frac{\epsilon_c^{\text{parent}}}{\epsilon_c^{\text{PW92}}} - 1,$$

the exchange rows posed per spin channel at the doubled density  $(2n_\sigma, 4\sigma_{\sigma\sigma})$  [OP79]. The objective is the integration-weighted pointwise residual plus a per-system energy term:

$$\mathcal{L}_{\text{pre}} = \frac{\sum_i w_i (F_i^{\text{NN}} - 1 - t_i)^2}{\sum_i w_i} + w_E \frac{1}{N_{\text{sys}}} \sum_s \left( \sum_{i \in s} w_i^{\text{grid}} n_i \epsilon_i^{\text{LDA}} F_i^{\text{NN}} - E_{xc,s}^{\text{parent}} \right)^2,$$
$$w_i = |n_i \epsilon_i^{\text{LDA}}| w_i^{\text{grid}}, \quad w_E = 0.1.$$

- The energy term is needed because the pointwise residual alone does not bound a system's energy.
- Adam, 20000 steps, learning rate  $10^{-3}$  to  $10^{-5}$  from the midpoint, clip 1.0, a 20 percent validation slice scored every 50 steps with patience 300 checks [cfg pretrain:].

### 3. Pre-training: the fidelity certificate and the GGA clones

The saved networks are re-evaluated [code: cluster/fidelity.py, grid\_config.py:287-312] on every system at the production basis and grid and compared with the parent:

$$\Delta E_{xc,s} = E_{xc,s}^{\text{NN}} - E_{xc,s}^{\text{parent}}, \quad \Delta AE = \Delta E_{xc,\text{mol}} - \sum_A \Delta E_{xc,A},$$

$$\max_{\text{atoms}} |\Delta E_{xc,A}| \leq 1.0 \text{ mHa}, \quad \text{mean}_{\text{mol}} |\Delta AE| \leq 1.0 \text{ kcal/mol}, \quad \max_{\text{mol}} |\Delta AE| \leq 2.0 \text{ kcal/mol}.$$

A failing certificate holds the group's training array; every trained cell starts from a network that reproduces its parent to within these tolerances.

| Architecture (parent PBE) | max atom (mHa) | mean $ \Delta AE $ | max $ \Delta AE $  | Verdict |
|---------------------------|----------------|--------------------|--------------------|---------|
| deep_2x8                  | 0.369          | 0.343              | 1.421 (C3H8, C4H6) | PASS    |
| deep_attn_2x8             | 0.201          | 0.131              | 0.573              | PASS    |
| deep_3x16                 | 0.210          | 0.162              | 0.519              | PASS    |
| deep_attn_3x16            | 0.058          | 0.031              | 0.115              | PASS    |
| deep0_3x16                | 0.439          | 0.114              | 0.450              | PASS    |
| deep0_attn_3x16           | 0.123          | 0.042              | 0.149              | PASS    |
| deep0_cusp_3x16           | 0.439          | 0.188              | 1.071 (AlCl3)      | PASS    |

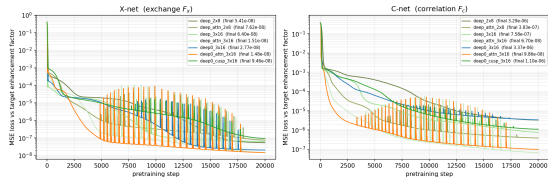
### 3. Pre-training: the meta-GGA clones

| Architecture (parent SCAN) | max atom (mHa) | mean $ \Delta AE $ | max $ \Delta AE $  | Verdict   |
|----------------------------|----------------|--------------------|--------------------|-----------|
| deep0_mgga_3x16            | 4.106          | 1.473              | 5.605 (11 species) | FAIL      |
| deep0_cusp_mgga_3x16       | 3.132          | 2.062              | 7.257 (14 species) | FAIL      |
| deep0_rung35_mgga_3x16     | 3.001          | 1.780              | 6.303 (11 species) | FAIL      |
| deep0_rung35ms_mgga_3x16   | 2.096          | 0.710              | 3.294 (4 species)  | FAIL      |
| deep0_mgga_attn_3x16       | -              | -                  | -                  | cancelled |

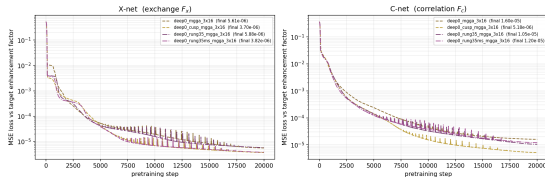
- Every completed SCAN clone fails on the free atoms (2.10 to 4.11 mHa against 1.0) and three of the four on the mean atomization error as well: the rung rather than one architecture is what fails at 3x16.
- Their exchange residuals end 40 to 400 times the GGA clones' and their correlation residuals 1.5 to 240 times, after the same 20000 steps (next slide).
- The meta-GGA training array is held; the lever study of Sec. 6 (wider networks, the energy-term weight) is the follow-up.

# 3. Pre-training: the loss curves

Pretraining loss curves -- 7 archs, 20000 steps  
run\_20260908T153908Z



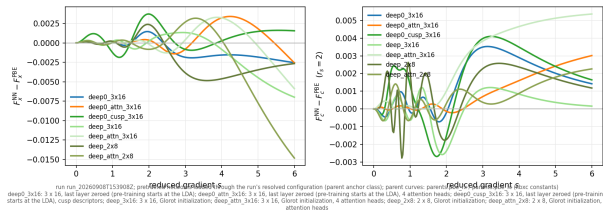
Pretraining loss curves -- 4 archs, 20000 steps  
run\_20260902T145250Z



- Pointwise residual per step, exchange (left panel of each pair) and correlation (right); left the seven certified GGA clones, right the four SCAN clones. The recurring bursts from step 4000 to 8000 on are a property of the fit, not of the validation, which reloads nothing.
- The GGA clones end at  $10^{-8}$  to  $10^{-6}$ ; deep\_attn\_3x16 descends earliest and lowest, at the longest fit (28 h against 3.8 h for its plain twin). The SCAN clones stop 40 to 400 times higher in exchange and fail the certificate.

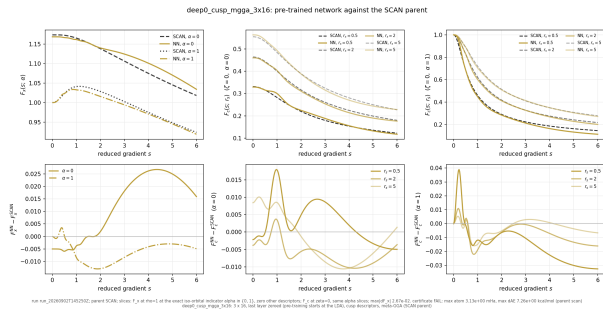
### 3. Pre-training: the GGA clones against PBE

Pre-trained networks against the PBE parent: differences, all architectures



- The difference  $F^{\text{NN}} - F^{\text{PBE}}$  of every GGA clone, exchange at  $n = 1$  and correlation at  $r_s = 2$ , along  $s$ ; one curve per architecture.
- Within  $2.5 \times 10^{-3}$  for  $s < 2.5$  ( $10^{-3}$  for the 3x16 clones), at most 0.015 at  $s = 6$  (deep\_attn\_2x8 exchange; 0.007 for deep\_3x16); the three zero-init clones within  $4.2 \times 10^{-3}$  everywhere.
- The region beyond  $s = 3$  is sparsely sampled by the pre-training set and is where the clones drift most; the training of Sec. 4 changes the functionals far more (0.1 to 1) than this.

### 3. Pre-training: a SCAN clone against SCAN



- One of the four SCAN clones (deep0\_cusp\_mgga\_3x16): the parent dashed, the network solid, the difference panels beside them, at  $r_s = 0.5, 2$  and  $5$  and at  $\alpha = 0$  and  $1$ .
- $F_x(\alpha = 0)$  is 0.01 to 0.027 above SCAN for  $s > 2.5$ ;  $F_c(\alpha = 1)$  0.0003 to 0.016 below it from  $s = 2$  at  $r_s = 2$  (0.005 to 0.033 at  $r_s = 0.5$ ).
- Same-sign offsets over wide  $s$  ranges do not cancel between a molecule and its atoms: this is why the clone fails on atomization energies while its pointwise residual is small. The rung-3.5 clones miss SCAN by 0.06 to 0.39 in correlation below  $s = 1$ ; deep0\_mgga\_3x16 sits within 0.01 on every slice and still fails on the atoms.

## 4. In-sample: the training objective [DFS21 Eqs. 15-17]

One optimizer step per training group per epoch (a reaction, or a molecule with its density and potential references); the group loss is the DFS weighting ( $\lambda_{RE} = 1$ ,  $\lambda_n = 20$ ), with  $E^{\text{NN}}$  the self-consistent energy of the  $N = 3$  cycle SCF and  $t$  the cycle:

$$\mathcal{L} = 1 \cdot \mathcal{L}_{AE} + 1 \cdot \mathcal{L}_{\text{rxn}} + 1 \cdot \mathcal{L}_{IP} + 1 \cdot \mathcal{L}_{V_{xc}} + 20 \cdot \mathcal{L}_{\rho},$$

$$\mathcal{L}_{\text{rxn}} = \frac{1}{|T|} \sum_{t \in T} w_t^2 r_t^2, \quad r_t = \sum_i c_i E_{i,t}^{\text{NN}} - E^{\text{ref}}, \quad w_t = \left( \frac{t}{N-1} \right)^2,$$

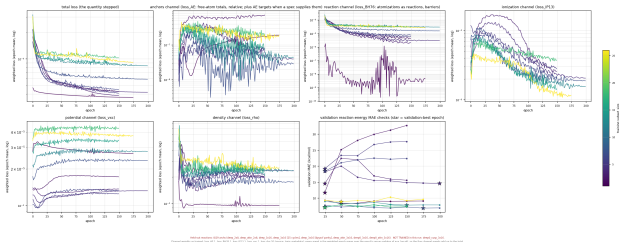
$$\mathcal{L}_{AE} = 0.01 \text{mean}_Z \frac{(E_Z^{\text{NN}} - E_Z^{\text{anchor}})^2}{(E_Z^{\text{anchor}})^2}, \quad \mathcal{L}_{IP} = \text{mean} (E_+^{\text{NN}} - E_0^{\text{NN}} - IP^{\text{ref}})^2, \quad \mathcal{L}_{V_{xc}} = \frac{\|V_{xc}^{\text{NN}} - V_{xc}^{\text{ref}}\|_F^2}{n_{\text{AO}}^2},$$

$$\mathcal{L}_{\rho} = \text{mean}_{\text{mol}} \frac{\sum_g w_g (n_g^{\text{NN}} - n_g^{\text{ref}})^2}{N_e^2}, \quad N_e = \sum_g w_g n_g^{\text{ref}}.$$

- $\mathcal{L}_{\text{rxn}}$  is the DFS convergence-tail scoring ( $w_t^2 = 0, 1/16, 1$  at  $N = 3$ ) over the W4-11 atomizations, as reactions with the network's own atoms, and the BH76 barriers [W411, GMTKN55]; the free atoms are held near their exact energies [Chak93];  $V_{xc}$  is compared on the reference density; the density against CCSD on the same grid, per electron. Code: losses.py:76, 366-396; train.py:1829-1835; oneshot.py:632-653.

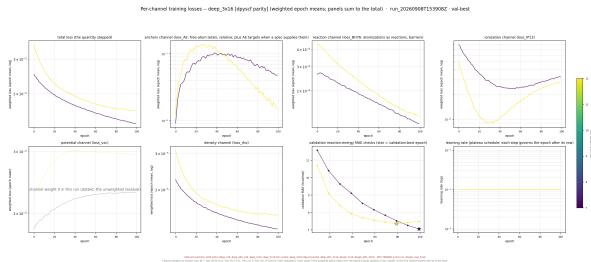
## 4. In-sample: the training losses (deep\_3x16)

Per-channel training losses -- deep\_3x16 (weighted epoch means; panels sum to the total) - run\_20200908T153908Z - val-best



- SCF: 3 cycles from the converged PBE density, DFS mixing  $D_{t+1} = (1 - a_t)D_t + a_tD_t^{\text{out}}$ ,  $a_t = 0.3^t + 0.3$  [DFS21 SI], gradients through every cycle, the last mixed density scored. AdamW  $10^{-3} \rightarrow 10^{-5}$  over the second half, weight decay  $10^{-4}$ , clip 1.0, 200 epochs.
- Seven panels, one curve per subset size: the total, the five channels at their trained weights, the validation error at each check. The reaction channel falls in every cell (by 2.3x to 2223x), the density channel by 1.15x to 4.6x, both settled within 25 epochs; the potential channel, absent from the DFS objective, rises in 43 of the 64 cells that carry it, falls in 9 and is flat in 12.

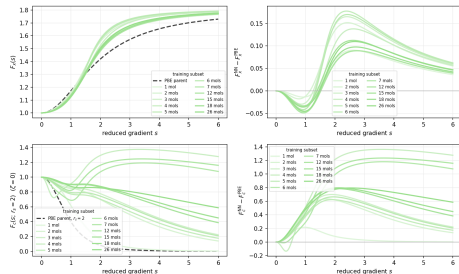
## 4. In-sample: validation and the arms' protocol



- Validation MAE on 35 reactions every 25 epochs picks the checkpoint. The 2-to-6-point cells of deep\_3x16, deep0\_3x16 and deep\_2x8 rise after the first check (deep\_3x16 at 2: 11.7 to 32.7 kcal/mol, best at epoch 25); the cells from seven points hold 4.7 to 11.6; the small cells of deep\_attn\_3x16 fall (37 to 7).
- The 25-cycle arm keeps everything else and runs 25 SCF cycles, 100 epochs, checks every 10. The parity arm (the figure, with its learning-rate panel) reproduces the DFS code's protocol: 25 cycles, the Letter's weights (reactions 0.01, no potential term), Adam with a plateau schedule at  $10^{-4}$  that never triggered.

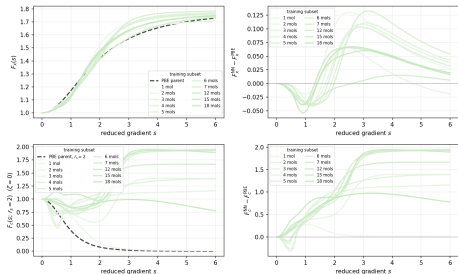
# 4. In-sample: the trained enhancement factors, the 3x16 networks

deep\_3x16: trained networks against the PBE parent



run: run\_3026090071530602; checkpoint char\_val\_best(model\_val\_best.apis); slices: F\_x at rho=1, pen-extra descriptors: F\_x at rho=0, r\_s=0; 11 completed cells; max|F\_x|\_1.13e-01; parent curves parents.pbe.fx / parents.pbe.fx; (blue constant); deep\_3x16: 3 x 16, Global initialization

deep\_attn\_3x16: trained networks against the PBE parent

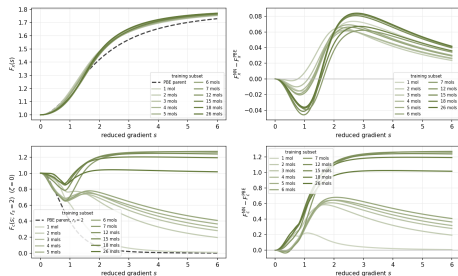


run: run\_3026090071530602; checkpoint char\_val\_best(model\_val\_best.apis); slices: F\_x at rho=1, pen-extra descriptors: F\_x at rho=0, r\_s=0; 10 completed cells; max|F\_x|\_1.13e-01; parent curves parents.pbe.fx / parents.pbe.fx; (blue constant); deep\_attn\_3x16: 3 x 16, Global initialization, 4 atbion heads

- Exchange: the trained  $F_x$  is below PBE by 0.002 to 0.06 near  $s = 0.8$  and above it from  $s = 1.0$  to 1.7, by up to 0.18 (deep\_3x16) and 0.13 (deep\_attn\_3x16) at  $s = 1.9$  to 3.4; the one-molecule cell of deep\_3x16 moves least.
- Correlation at  $r_s = 2$ : the one-molecule cells fall to 0.16 by  $s = 2$ ; the others stay 1 to  $s = 2$  (PBE 0.02 by  $s = 3$ ) and then split, deep\_3x16 at 2, 3, 5, 6, 15, 18, 26 decaying to 0.1 to 0.6 by  $s = 6$  while 4, 7 and 12 rise to 1.1 to 1.3, and every deep\_attn\_3x16 cell but the one-molecule and the eighteen-point ones rising above 1 (to 1.95 at  $s = 6$ ).

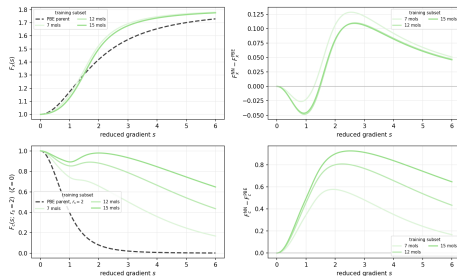
# 4. In-sample: the trained enhancement factors, the 2x8 pair and the arms

deep\_2x8: trained networks against the PBE parent



run run\_3036908071536602; checkpoint charnel\_val\_best(model\_val\_best.exp); slices:  $F_c$  at rho=1, par extra description:  $F_c$  at eta=0,  $r_p=0$ ; 11 completed cells; model@\_x 8.40e-02; parent curves parents.pbe,  $F_c$  / parents.pbe to: (blank constant)

deep\_3x16 [25 cycles]: trained networks against the PBE parent

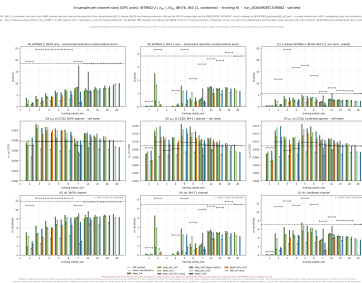


run run\_3036908071536602; checkpoint charnel\_val\_best(model\_val\_best.exp); slices:  $F_c$  at rho=1, par extra description:  $F_c$  at eta=0,  $r_p=0$ ; 3 completed cells; model@\_x 1.20e-02; parent curves parents.pbe,  $F_c$  / parents.pbe to: (blank constant)

deep\_3x16 [25 cycles] 3 x 16, Global initialization, 25 cycles

- Left, deep\_2x8: the same exchange shape (above PBE by up to 0.08 at  $s = 1.9$  to 3.4; deep\_attn\_2x8 by 0.19); in correlation its cells from seven points rise to 1.0 to 1.3 while its small cells decay to 0.2 to 0.4 by  $s = 6$ .
- Right, the 25-cycle arm at 7 and 12: exchange above PBE by up to 0.13, as the parity arm; the parity arm's  $F_c$  stays within 0.08 of PBE on every drawn slice (the Letter's weights leave the correlation factor near its start), the 25-cycle arm's within the deep\_3x16 spread.

## 4. In-sample: energy and density in DFS units on the training reactions



- Rows WTMA2-2 /  $\epsilon_n$  /  $\mathcal{ED}$ , columns BH76 / W4-11 / combined; bars against PBE on each cell's own training reactions (caps), the dashed line PBE on their union (18.6, 3.83, 5.67 kcal/mol).
- All 66 trained cells (the two set-aside cells included) are below PBE on  $\mathcal{ED}$  over their own reactions: training WTMA2-2 0.00 to 5.4 kcal/mol against 5.67 on the union.
- The grid-RMSE density of the trained molecules is below PBE in every cell (NN/PBE 0.57 to 0.97 over the architectures); the per-electron  $\epsilon_n$  of deep\_3x16 is above the pooled PBE value from 2 to 18 points, so the two density measures disagree on the small cells.

## 5. Held-out: the set and the metrics [Goe17; DFS21 Eqs. 20, 21]

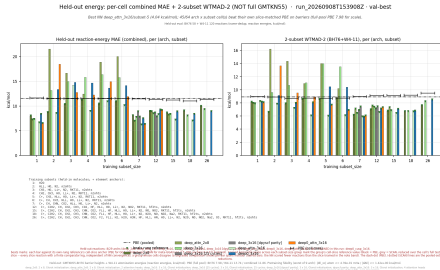
Test set: GMTKN55 BH76 barriers and W4-11 atomization energies, minus the 35-reaction validation slice and each cell's training reactions: 61 BH76 rows (three listed twice by name, four more as permuted-name twins; 54 barrier identities are scored) and 120 W4-11 reactions, 189 species with CCSD densities; the tables' n rxn column counts name-deduplicated rows (58 BH76 at one point). PBE is evaluated on each cell's own slice; the functionals with the same 3-cycle SCF they trained with, PBE's density converged.

$$\text{WTMAD-2} = \frac{56.84}{N} \sum_{i \in \{\text{BH76}, \text{W4-11}\}} N_i \frac{\text{MAD}_i}{|\Delta E_{\text{ref}}|_i}, \quad \varepsilon_n = \frac{1}{N_e} \int |n - n^{\text{ref}}| d^3r, \quad N_e = \int n^{\text{ref}} d^3r,$$

$$\mathcal{ED} = \frac{2}{1/E + 1/(\gamma D)}, \quad D_{\text{RMSE}} = \sqrt{\sum_g w_g (n_g - n_g^{\text{ref}})^2 / \sum_g w_g}.$$

- WTMAD-2 over the two present subsets; alone, a pool reads  $56.84 \text{ MAD}_i / |\Delta E_{\text{ref}}|_i$ , a scaled relative error.  $\gamma = 1084.87 \text{ kcal/mol}$  on  $\varepsilon_n$  (DFS units) or  $\gamma = E_{\text{PBE}}/D_{\text{PBE}}$  on the grid RMSE (self-calibrated).
- A cell beats PBE when its  $\mathcal{ED}$  is below PBE's on the same slice. Pooled PBE: 8.92 kcal/mol WTMAD-2,  $\varepsilon_n$  0.00923.
- The harmonic mean is set by its smaller leg: with  $\gamma \varepsilon_n$  at 9.6 to 13.6 kcal/mol here, 1 kcal/mol of  $E$  moves  $\mathcal{ED}$  by 0.4 to 0.9, the whole density spread by at most 2.7 (1.3 at  $E = 8$ ).

## 5. Held-out: energy

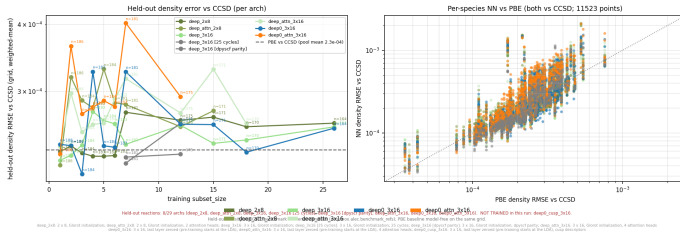


- Combined WTMAD-2 against subset size, one line per architecture, PBE 8.92 dashed. deep\_3x16 7.92 at one point, 9.61 to 13.97 at two to six, 6.44 to 7.41 from seven to eighteen, 8.23 at twenty-six; deep0\_3x16 (5.90 to 10.48) below PBE at 1, 3, 4 and from seven, above at 2, 5 and 6.
- deep\_attn\_3x16 (5.94 to 8.26) and deep\_2x8 (6.18 to 8.85) are below PBE at every cell; deep0\_attn\_3x16 below at 1 and 4 to 12 (6.08 to 8.87), above at 2 and 3; deep\_attn\_2x8 below at 1, 7, 12 and 15, above at 2 to 6 (10.19 to 16.19).
- The arms at 7 and 12: 6.92 and 7.17 (25 cycles), 7.44 and 7.48 (parity). The largest subsets (18, 26) do not improve on 7 to 15.

# 5. Held-out: density

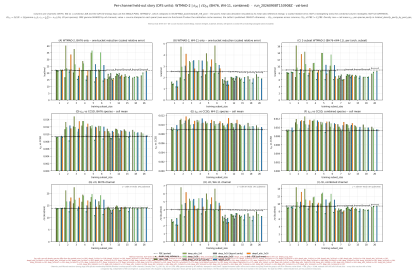
Held-out density error vs CCSD (NN vs PBE) · run\_20260908T153908Z · val-best

HELD-OUT density error vs CCSD reference densities on the W4-11+8H76 benchmark species (atoms excluded; weighted-mean grid RMSE, NOT  $N_e$ -normalized). PBE baseline is model-free on the same grid. Density generalization; separate from the held-out ENERGY panels by design. Held-out val: 101% 10 = 100-11 120 reactions (train+dev; reaction energies, kcal/mol); density: 149 species vs CCSD refs at matching basisset (atoms excluded).



- The held-out density error against CCSD, per cell, beside PBE's on the same species: in the per-electron measure  $\varepsilon_n$  (PBE 0.00923) the trained functionals are at or above PBE in every cell but five (deep\_attn\_2x8 at one point, deep\_attn\_3x16 at five, deep0\_attn\_3x16 at one, the parity arm at 7 and 12), by 0.2 to 36 percent.
- A fixed set of eight species accounts for the excess (the tail, two slides on); the comparison is asymmetric by protocol: three SCF cycles for the network against a converged PBE density.

## 5. Held-out: energy and density in DFS units (full set)



- Rows WTMAD-2 /  $\epsilon_n$  /  $\mathcal{ED}$ , columns BH76 / W4-11 / combined; the dashed line PBE on the pooled slice, the black cap PBE on each cell's own slice, a triangle a cell below its cap, a star the SCAN comparator where it exists; the  $\gamma$  stamp names the units.
- $\mathcal{ED}$ , 41 of the 64 cells below PBE: deep\_3x16 and deep0\_3x16 at 1, 7, 12, 15, 18, 26 (the two starts agree at every size); deep\_attn\_3x16 at every cell (lowest 7.40 to 7.85 at 4 to 7 points against caps of 9.32 to 9.49); deep0\_attn\_3x16 at 1, 6, 7, 12; deep\_2x8 at 1, 2 and 7 to 26; deep\_attn\_2x8 at 1, 7, 12, 15; both arms at 7 and 12.
- The  $\mathcal{ED}$  verdict follows the energy verdict in 56 of the 64 cells; in eight (deep\_2x8 at 3 to 6, deep0\_3x16 at 3 and 4, deep0\_attn\_3x16 at 4 and 5) the energy leg is below PBE and the density leg lifts  $\mathcal{ED}$  above it.

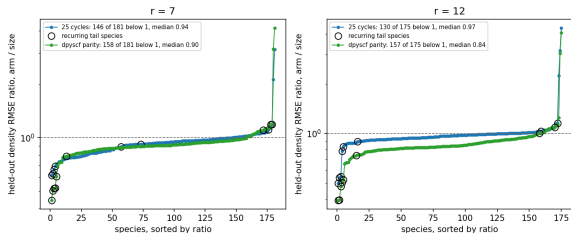


## 5. The density program: 25 cycles against 3, the same architecture and subsets

| Cell                      | $r$ | WTMAD-2 | $\varepsilon_n$ | grid RMSE | $\mathcal{ED}$ | $\mathcal{ED}$ PBE cap |
|---------------------------|-----|---------|-----------------|-----------|----------------|------------------------|
| deep_3x16                 | 7   | 6.44    | 0.01017         | 2.39e-04  | 8.13           | 9.49                   |
| deep_3x16 [25 cycles]     | 7   | 6.92    | 0.00972         | 2.20e-04  | 8.36           | 9.49                   |
| deep_3x16 [dpyscf parity] | 7   | 7.44    | 0.00903         | 2.26e-04  | 8.46           | 9.49                   |
| deep_3x16                 | 12  | 7.41    | 0.01047         | 2.59e-04  | 8.97           | 9.56                   |
| deep_3x16 [25 cycles]     | 12  | 7.17    | 0.00983         | 2.62e-04  | 8.57           | 9.56                   |
| deep_3x16 [dpyscf parity] | 12  | 7.48    | 0.00892         | 2.29e-04  | 8.44           | 9.56                   |
| PBE                       | -   | 8.92    | 0.00923         | 2.34e-04  | -              | -                      |

- Yes on density, modestly: the 25-cycle arm lowers  $\varepsilon_n$  by 4.3 percent at 7 and 6.2 percent at 12 and stays above PBE; in the grid RMSE the gain holds at 7 and vanishes at 12 (o3, c2 and fo2 regress). On energy it is a wash and every cell stays under its PBE cap.
- The parity arm is the density win:  $\varepsilon_n$  0.00903 and 0.00892, the only deep\_3x16 cells below PBE, bought with a worse energy leg (BH76 WTMAD-2 19.5 against 16.3 and 18.9).

## 5. The density program: per species



val\_best\_channel: each cell evaluated with its own SCF cycle budget (density species): size cell at  $r = 7$ : max\_cycles 3, 1 of 181 converged; size cell at  $r = 12$ : max\_cycles 3, 0 of 175 converged; 25 cycles at  $r = 7$ : max\_cycles 25, 168 of 181 converged; dpyscf parity at  $r = 7$ : max\_cycles 25, 173 of 181 converged; 25 cycles at  $r = 12$ : max\_cycles 25, 161 of 175 converged; dpyscf parity at  $r = 12$ : max\_cycles 25, 169 of 175 converged.

- Per held-out species the ratio of the arm's density RMSE to the size cell's, sorted; below 1 is an improvement. The 25-cycle arm improves 146 of 181 species at 7 (median ratio 0.94) and 130 of 175 at 12 (0.97); the parity arm 158 of 181 (0.90) and 157 of 175 (0.84).
- The species helped most (the left ends) are the recurring tail, drawn as hollow markers: part of the tail is the three-cycle evaluation rather than the functionals.
- Caveat: the arms were also *evaluated* with 25 cycles (161 to 173 of the species converged) and the size cells with 3 (1 and 0 converged), so training and evaluation protocol are conflated; the converged-SCF re-evaluation of the same networks is the direct test.

## 5. Held-out: per-cell verdicts, combined leg, DFS units

| Architecture              | Below PBE | At $r =$                |
|---------------------------|-----------|-------------------------|
| deep_3x16                 | 6 of 11   | 1, 7, 12, 15, 18, 26    |
| deep_attn_3x16            | 10 of 10  | every cell              |
| deep0_3x16                | 6 of 11   | 1, 7, 12, 15, 18, 26    |
| deep0_attn_3x16           | 4 of 8    | 1, 6, 7, 12             |
| deep_2x8                  | 7 of 11   | 1, 2, 7, 12, 15, 18, 26 |
| deep_attn_2x8             | 4 of 9    | 1, 7, 12, 15            |
| deep_3x16 [25 cycles]     | 2 of 2    | 7, 12                   |
| deep_3x16 [dpyscf parity] | 2 of 2    | 7, 12                   |

- 41 of the 64 cells; the numbers behind every row are on the backup tables. Verdicts within about 10 percent of PBE on a leg depend on the leg and the units: ten cells flip under the plain-MAE leg or the self-calibrated grid-RMSE units, the other 54 carry the same verdict under every choice.

[▶ 3x16 table](#)[▶ 2x8 and arms](#)[▶ the flips](#)

## 6. Pending: running, queued and set aside

- Cells running or queued: `deep_attn_3x16` at 26 (stopped by the 48 h wall, relaunched from its resume checkpoint), `deep_attn_2x8` at 26, `deep0_attn_3x16` at 15 to 26, `deep0_cusp_3x16` at every subset, the arms at 15 to 26.
- Two evaluated cells (`deep_attn_2x8` at 18, `deep_3x16` [25 cycles] at 15) await re-evaluation: their passes ran while the T1 backfill was rewriting the reference files with a key the reference reader did not accept, so every reference was refused and the tables held no reaction.
- The corrected reader (the key accepted; an evaluation that reads no species refused instead of written) is committed and is deployed with the re-runs in a later cluster session; every task whose evaluation starts before that deploy writes the same tables.
- The figures and tables regenerate from the same suite as cells land, by one entry per run in `family_runs.yaml`.

## 6. Pending: the open decisions

- Meta-GGA group: four certificates FAIL, one fit cancelled; the training array is held. Landed follow-up: three wider architectures (3x32, 4x16, 4x32) registered and a lever study (fourteen 10000-step fits: energy-term weight 0.1, 10, 100 over the four widths, plus the integration objective on the 3x16) as one 48 h job awaiting submission.
- Converged-SCF re-evaluation: one cell-pass (deep\_3x16 at 1, final checkpoint, 214 of 214 converged) took about 13 h of a 24 h job; the campaign at two checkpoints is about 1700 node-hours, so the program's shape is a decision still open. It is the direct test behind the density program's numbers.
- The T1 diagnostics landed (203 species, threshold 0.02); the model-free tail-excluded views are rendered from them.
- The figure suite's in-sample atomization-energy panel stays out of the documents until it uses the network's own atoms (it now subtracts fixed exact atomic energies and reports the absolute-energy offset).

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## Backup: the runs

| Run               | Architectures                                                                                                 | Cells done / planned                                                                | Run directory        |
|-------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------|
| size              | deep_3x16, deep_attn_3x16, deep_2x8, deep_attn_2x8                                                            | deep_3x16 11/11,<br>deep_attn_3x16 10/11,<br>deep_2x8 11/11,<br>deep_attn_2x8 9/11  | run_20260902T145245Z |
| families          | deep0_3x16, deep0_cusp_3x16                                                                                   | deep0_attn_3x16, deep0_3x16 11/11,<br>deep0_attn_3x16 8/11,<br>deep0_cusp_3x16 0/11 | run_20260902T145247Z |
| meta-GGA families | deep0_cusp_mgga_3x16, deep0_mgga_3x16, deep0_mgga_attn_3x16, deep0_rung35_mgga_3x16, deep0_rung35ms_mgga_3x16 | 0/55 (four certificates FAIL, one fit cancelled)                                    | run_20260902T145250Z |
| 25-cycle arm      | deep_3x16 [25 cycles]                                                                                         | 2/5 (7, 12)                                                                         | run_20260908T153856Z |
| dpyscf-parity arm | deep_3x16 [dpyscf parity]                                                                                     | 2/5 (7, 12)                                                                         | run_20260908T153908Z |

[◀ back to Scope](#)

# Backup: the eleven architectures

| Architecture                             | Last layer | Rung, parent   | Hidden | Heads | Extra                              |
|------------------------------------------|------------|----------------|--------|-------|------------------------------------|
| deep_2x8                                 | Glorot     | GGA, PBE       | 2 x 8  | none  | none                               |
| deep_attn_2x8                            | Glorot     | GGA, PBE       | 2 x 8  | 2     | none                               |
| deep_3x16                                | Glorot     | GGA, PBE       | 3 x 16 | none  | none                               |
| deep_attn_3x16                           | Glorot     | GGA, PBE       | 3 x 16 | 4     | none                               |
| deep0_3x16                               | zeroed     | GGA, PBE       | 3 x 16 | none  | none                               |
| deep0_attn_3x16                          | zeroed     | GGA, PBE       | 3 x 16 | 4     | none                               |
| deep0_cusp_3x16                          | zeroed     | GGA, PBE       | 3 x 16 | none  | cusp                               |
| deep0_mgga_3x16 (_attn: 4 heads)         | zeroed     | meta-GGA, SCAN | 3 x 16 | none  | $\alpha$                           |
| deep0_cusp_mgga_3x16                     | zeroed     | meta-GGA, SCAN | 3 x 16 | none  | cusp, $\alpha$                     |
| deep0_rung35_mgga_3x16 (_ms: multishell) | zeroed     | meta-GGA, SCAN | 3 x 16 | none  | cusp, rung-3.5 occupancy, $\alpha$ |

Inputs:  $F_x$  from  $x_s$ ,  $F_c$  from  $x_0, x_1, x_s$ ; the cusp architectures add  $x_4, x_5$  to both, the meta-GGA ones  $x_\alpha$  as well, the rung-3.5 ones their occupancy columns (\_ms: one column per shell). The arms' cells carry the run's tag on the same axis: deep\_3x16 [25 cycles], deep\_3x16 [dpyscf parity].

[◀ back to the architectures](#)

# Backup: the 26-point DFS pool

| Idx | Point | Species (name, charge, spin) | Idx | Point                  | Species (name, charge, spin)                     |
|-----|-------|------------------------------|-----|------------------------|--------------------------------------------------|
| 0   | H2    | H2 (0,0), H (0,1)            | 13  | NO2                    | NO2 (0,1)                                        |
| 1   | N2    | N2 (0,0)                     | 14  | HN                     | HN (0,2), H (0,1)                                |
| 2   | FLi   | FLi (0,0), Li (0,1)          | 15  | O3                     | O3 (0,0)                                         |
| 3   | CHN   | CHN (0,0), H (0,1)           | 16  | N2O                    | N2O (0,0)                                        |
| 4   | CO2   | CO2 (0,0)                    | 17  | CH3                    | CH3 (0,1), H (0,1)                               |
| 5   | F2    | F2 (0,0)                     | 18  | CH2                    | CH2 (0,2), H (0,1)                               |
| 6   | C2H2  | C2H2 (0,0), H (0,1)          | 19  | H2O                    | H2O (0,0), H (0,1)                               |
| 7   | CO    | CO (0,0)                     | 20  | H3N                    | H3N (0,0), H (0,1)                               |
| 8   | HLi   | HLi (0,0), H (0,1), Li (0,1) | 21  | OH+N2 to H+N2O (BH76)  | HO (0,1), N2 (0,0), H (0,1), N2O (0,0)           |
| 9   | Na2   | Na2 (0,0)                    | 22  | OH+CH3 to O+CH4 (BH76) | HO (0,1), CH3 (0,1), O (0,2), CH4 (0,0), H (0,1) |
| 10  | NO    | NO (0,1)                     | 23  | HF+F to H+F2 (BH76)    | HF (0,0), F (0,1), H (0,1), F2 (0,0)             |
| 11  | CH    | CH (0,1), H (0,1)            | 24  | Li IP (IP13)           | Li (0,1), Li+ (1,0)                              |
| 12  | HO    | HO (0,1), H (0,1)            | 25  | C IP (IP13)            | C (0,2), C+ (1,1)                                |

References: atomization energies [HK12] with H2O and C2H2 from [W411]; reaction energies [Goe17, BH76RC]; ionization potentials [NIST]. Code: dfs\_pool.py, training\_points.py:347-420.

[◀ back to the pool](#)

## Backup: the eleven chosen subsets

| <i>r</i> Points (ledger index order)                                                                                                                                                                                                                                                               | AE / BH76 / IP | JSD    | $\rho^{1/3}$ | <i>s</i> | $\alpha$ |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------|--------------|----------|----------|
| 1 H <sub>2</sub> O                                                                                                                                                                                                                                                                                 | 1 / 0 / 0      | 0.0475 | 0.00109      | 0.03764  | 0.00881  |
| 2 HLi, OH+N <sub>2</sub> to H+N <sub>2</sub> O                                                                                                                                                                                                                                                     | 1 / 1 / 0      | 0.0284 | 0.00015      | 0.02372  | 0.00455  |
| 3 OH+N <sub>2</sub> to H+N <sub>2</sub> O, OH+CH <sub>3</sub> to O+CH <sub>4</sub> , Li IP                                                                                                                                                                                                         | 0 / 2 / 1      | 0.0228 | 0.00017      | 0.01995  | 0.00267  |
| 4 CH <sub>2</sub> , OH+N <sub>2</sub> to H+N <sub>2</sub> O, OH+CH <sub>3</sub> to O+CH <sub>4</sub> , Li IP                                                                                                                                                                                       | 1 / 2 / 1      | 0.0203 | 0.00008      | 0.01801  | 0.00218  |
| 5 HLi, OH+N <sub>2</sub> to H+N <sub>2</sub> O, OH+CH <sub>3</sub> to O+CH <sub>4</sub> , Li IP, C IP                                                                                                                                                                                              | 1 / 2 / 2      | 0.0163 | 0.00028      | 0.01418  | 0.00183  |
| 6 HLi, CH, OH+N <sub>2</sub> to H+N <sub>2</sub> O, OH+CH <sub>3</sub> to O+CH <sub>4</sub> , Li IP, C IP                                                                                                                                                                                          | 2 / 2 / 2      | 0.0130 | 0.00021      | 0.01120  | 0.00156  |
| 7 CHN, CO <sub>2</sub> , HLi, CH, OH+N <sub>2</sub> to H+N <sub>2</sub> O, Li IP, C IP                                                                                                                                                                                                             | 4 / 1 / 2      | 0.0102 | 0.00027      | 0.00877  | 0.00120  |
| 12 CO <sub>2</sub> , C <sub>2</sub> H <sub>2</sub> , CO, HLi, CH, N <sub>2</sub> O, CH <sub>2</sub> , OH+N <sub>2</sub> to H+N <sub>2</sub> O, OH+CH <sub>3</sub> to O+CH <sub>4</sub> , HF+F to H+F <sub>2</sub> , Li IP, C IP                                                                    | 7 / 3 / 2      | 0.0034 | 0.00004      | 0.00292  | 0.00040  |
| 15 FLi, CHN, CO <sub>2</sub> , C <sub>2</sub> H <sub>2</sub> , CO, HLi, NO, CH, NO <sub>2</sub> , CH <sub>2</sub> , OH+N <sub>2</sub> to H+N <sub>2</sub> O, OH+CH <sub>3</sub> to O+CH <sub>4</sub> , HF+F to H+F <sub>2</sub> , Li IP, C IP                                                      | 10 / 3 / 2     | 0.0010 | 0.00007      | 0.00081  | 0.00017  |
| 18 N <sub>2</sub> , FLi, CHN, CO <sub>2</sub> , C <sub>2</sub> H <sub>2</sub> , CO, HLi, Na <sub>2</sub> , NO, CH, NO <sub>2</sub> , N <sub>2</sub> O, CH <sub>2</sub> , OH+N <sub>2</sub> to H+N <sub>2</sub> O, OH+CH <sub>3</sub> to O+CH <sub>4</sub> , HF+F to H+F <sub>2</sub> , Li IP, C IP | 13 / 3 / 2     | 0.0004 | 0.00003      | 0.00028  | 0.00007  |
| 26 the whole pool                                                                                                                                                                                                                                                                                  | 21 / 3 / 2     | 0      | 0            | 0        | 0        |

[◀ back to the chosen subsets](#)

## Backup: the fidelity certificate of every clone

| Architecture             | Parent | Loss X / C       | max atom (mHa) | mean $ \Delta AE $ | max $ \Delta AE $  | Verdict |
|--------------------------|--------|------------------|----------------|--------------------|--------------------|---------|
| deep_2x8                 | PBE    | 5.4e-8 / 3.3e-6  | 0.369          | 0.343              | 1.421 (C3H8, C4H6) | PASS    |
| deep_attn_2x8            | PBE    | 7.6e-8 / 3.8e-7  | 0.201          | 0.131              | 0.573              | PASS    |
| deep_3x16                | PBE    | 6.4e-8 / 7.6e-7  | 0.210          | 0.162              | 0.519              | PASS    |
| deep_attn_3x16           | PBE    | 1.5e-8 / 6.7e-8  | 0.058          | 0.031              | 0.115              | PASS    |
| deep0_3x16               | PBE    | 2.8e-8 / 3.4e-6  | 0.439          | 0.114              | 0.450              | PASS    |
| deep0_attn_3x16          | PBE    | 1.5e-8 / 9.9e-8  | 0.123          | 0.042              | 0.149              | PASS    |
| deep0_cusp_3x16          | PBE    | 9.5e-8 / 1.1e-6  | 0.439          | 0.188              | 1.071 (AlCl3)      | PASS    |
| deep0_mgga_3x16          | SCAN   | 5.6e-6 / 1.6e-5  | 4.106          | 1.473              | 5.605 (11 species) | FAIL    |
| deep0_cusp_mgga_3x16     | SCAN   | 3.7e-6 / 5.2e-6  | 3.132          | 2.062              | 7.257 (14 species) | FAIL    |
| deep0_rung35_mgga_3x16   | SCAN   | 5.9e-6 / 1.05e-5 | 3.001          | 1.780              | 6.303 (11 species) | FAIL    |
| deep0_rung35ms_mgga_3x16 | SCAN   | 3.8e-6 / 1.2e-5  | 2.096          | 0.710              | 3.294 (4 species)  | FAIL    |

The pointwise loss is the final value of the pre-training residual for the exchange (X) and correlation (C) networks; the species in parentheses are the molecules above 1 kcal/mol.

[◀ back to the certificate](#)

# Backup: per-cell results, combined leg, DFS units, the 3x16 families [holdout\_by\_pool\_3x3\_eps.csv]

| Arch           | r  | E NN  | E PBE | eps NN  | eps PBE | ED NN | ED PBE | $\Delta$ ED  | Arch            | r  | E NN  | E PBE | eps NN  | eps PBE | ED NN | ED PBE | $\Delta$ ED  |
|----------------|----|-------|-------|---------|---------|-------|--------|--------------|-----------------|----|-------|-------|---------|---------|-------|--------|--------------|
| deep_3x16      | 1  | 7.92  | 8.92  | 0.00951 | 0.00923 | 8.96  | 9.46   | <b>-0.50</b> | deep_attn_3x16  | 18 | 6.72  | 8.92  | 0.00993 | 0.00923 | 8.27  | 9.62   | <b>-1.35</b> |
| deep_3x16      | 2  | 9.61  | 8.92  | 0.01071 | 0.00923 | 10.52 | 9.44   | +1.08        | deep0_3x16      | 1  | 8.18  | 8.92  | 0.00997 | 0.00923 | 9.32  | 9.46   | <b>-0.14</b> |
| deep_3x16      | 3  | 10.67 | 8.92  | 0.01068 | 0.00923 | 11.11 | 9.32   | +1.79        | deep0_3x16      | 2  | 9.16  | 8.92  | 0.01143 | 0.00923 | 10.54 | 9.44   | +1.10        |
| deep_3x16      | 4  | 11.06 | 8.92  | 0.01182 | 0.00923 | 11.88 | 9.34   | +2.53        | deep0_3x16      | 3  | 7.99  | 8.92  | 0.01062 | 0.00923 | 9.44  | 9.32   | +0.12        |
| deep_3x16      | 5  | 13.97 | 8.92  | 0.01113 | 0.00923 | 12.96 | 9.32   | +3.64        | deep0_3x16      | 4  | 8.06  | 8.92  | 0.01160 | 0.00923 | 9.83  | 9.34   | +0.48        |
| deep_3x16      | 6  | 13.51 | 8.92  | 0.01091 | 0.00923 | 12.62 | 9.34   | +3.28        | deep0_3x16      | 5  | 10.48 | 8.92  | 0.01099 | 0.00923 | 11.16 | 9.32   | +1.83        |
| deep_3x16      | 7  | 6.44  | 8.92  | 0.01017 | 0.00923 | 8.13  | 9.49   | <b>-1.35</b> | deep0_3x16      | 6  | 10.40 | 8.92  | 0.01097 | 0.00923 | 11.10 | 9.34   | +1.76        |
| deep_3x16      | 12 | 7.41  | 8.92  | 0.01047 | 0.00923 | 8.97  | 9.56   | <b>-0.59</b> | deep0_3x16      | 7  | 5.90  | 8.92  | 0.01051 | 0.00923 | 7.77  | 9.49   | <b>-1.72</b> |
| deep_3x16      | 15 | 6.89  | 8.92  | 0.01004 | 0.00923 | 8.44  | 9.60   | <b>-1.16</b> | deep0_3x16      | 12 | 7.11  | 8.92  | 0.01058 | 0.00923 | 8.78  | 9.56   | <b>-0.78</b> |
| deep_3x16      | 18 | 6.74  | 8.92  | 0.00999 | 0.00923 | 8.31  | 9.62   | <b>-1.31</b> | deep0_3x16      | 15 | 7.10  | 8.92  | 0.01013 | 0.00923 | 8.63  | 9.60   | <b>-0.97</b> |
| deep_3x16      | 26 | 8.23  | 8.92  | 0.00973 | 0.00923 | 9.25  | 9.80   | <b>-0.55</b> | deep0_3x16      | 18 | 6.82  | 8.92  | 0.00996 | 0.00923 | 8.37  | 9.62   | <b>-1.26</b> |
| deep_attn_3x16 | 1  | 8.26  | 8.92  | 0.00925 | 0.00923 | 9.06  | 9.46   | <b>-0.40</b> | deep0_3x16      | 26 | 8.52  | 8.92  | 0.00961 | 0.00923 | 9.38  | 9.80   | <b>-0.42</b> |
| deep_attn_3x16 | 2  | 7.87  | 8.92  | 0.00972 | 0.00923 | 9.01  | 9.44   | <b>-0.43</b> | deep0_attn_3x16 | 1  | 8.11  | 8.92  | 0.00906 | 0.00923 | 8.89  | 9.46   | <b>-0.57</b> |
| deep_attn_3x16 | 3  | 7.22  | 8.92  | 0.00981 | 0.00923 | 8.60  | 9.32   | <b>-0.72</b> | deep0_attn_3x16 | 2  | 13.64 | 8.92  | 0.01168 | 0.00923 | 13.14 | 9.44   | +3.70        |
| deep_attn_3x16 | 4  | 6.09  | 8.92  | 0.01020 | 0.00923 | 7.85  | 9.34   | <b>-1.49</b> | deep0_attn_3x16 | 3  | 9.52  | 8.92  | 0.01177 | 0.00923 | 10.91 | 9.32   | +1.59        |
| deep_attn_3x16 | 5  | 6.03  | 8.92  | 0.00883 | 0.00923 | 7.40  | 9.32   | <b>-1.92</b> | deep0_attn_3x16 | 4  | 8.87  | 8.92  | 0.01127 | 0.00923 | 10.28 | 9.34   | +0.93        |
| deep_attn_3x16 | 6  | 6.09  | 8.92  | 0.00973 | 0.00923 | 7.73  | 9.34   | <b>-1.61</b> | deep0_attn_3x16 | 5  | 7.73  | 8.92  | 0.01134 | 0.00923 | 9.49  | 9.32   | +0.17        |
| deep_attn_3x16 | 7  | 5.94  | 8.92  | 0.00963 | 0.00923 | 7.57  | 9.49   | <b>-1.91</b> | deep0_attn_3x16 | 6  | 6.90  | 8.92  | 0.01078 | 0.00923 | 8.68  | 9.34   | <b>-0.66</b> |
| deep_attn_3x16 | 12 | 6.61  | 8.92  | 0.01019 | 0.00923 | 8.27  | 9.56   | <b>-1.29</b> | deep0_attn_3x16 | 7  | 6.08  | 8.92  | 0.01126 | 0.00923 | 8.12  | 9.49   | <b>-1.37</b> |
| deep_attn_3x16 | 15 | 6.47  | 8.92  | 0.00956 | 0.00923 | 7.97  | 9.60   | <b>-1.63</b> | deep0_attn_3x16 | 12 | 7.34  | 8.92  | 0.01100 | 0.00923 | 9.09  | 9.56   | <b>-0.47</b> |

E in kcal/mol (2-subset WTAD-2), eps per electron, ED under  $\gamma = 1084.87$  kcal/mol; E PBE and eps PBE are the pooled values, ED PBE the cell-slice value.  $\Delta$ ED = ED NN – ED PBE: **bold, negative** marks a cell below PBE (26 of these 40 cells).

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# Backup: per-cell results, combined leg, DFS units, the 2x8 pair and the arms [holdout\_by\_pool\_3x3\_eps.csv]

| Arch                  | <i>r</i> | E NN | E PBE | eps NN  | eps PBE | ED NN | ED PBE | ΔED          | Arch                      | <i>r</i> | E NN  | E PBE | eps NN  | eps PBE | ED NN | ED PBE | ΔED          |
|-----------------------|----------|------|-------|---------|---------|-------|--------|--------------|---------------------------|----------|-------|-------|---------|---------|-------|--------|--------------|
| deep_2x8              | 1        | 8.18 | 8.92  | 0.00952 | 0.00923 | 9.13  | 9.46   | <b>-0.33</b> | deep_attn_2x8             | 1        | 7.98  | 8.92  | 0.00909 | 0.00923 | 8.82  | 9.46   | <b>-0.64</b> |
| deep_2x8              | 2        | 6.65 | 8.92  | 0.01116 | 0.00923 | 8.58  | 9.44   | <b>-0.85</b> | deep_attn_2x8             | 2        | 16.19 | 8.92  | 0.01193 | 0.00923 | 14.39 | 9.44   | +4.95        |
| deep_2x8              | 3        | 7.92 | 8.92  | 0.01148 | 0.00923 | 9.69  | 9.32   | +0.37        | deep_attn_2x8             | 3        | 14.36 | 8.92  | 0.01254 | 0.00923 | 13.97 | 9.32   | +4.65        |
| deep_2x8              | 4        | 8.62 | 8.92  | 0.01152 | 0.00923 | 10.20 | 9.34   | +0.86        | deep_attn_2x8             | 4        | 10.97 | 8.92  | 0.01156 | 0.00923 | 11.70 | 9.34   | +2.36        |
| deep_2x8              | 5        | 8.58 | 8.92  | 0.01128 | 0.00923 | 10.09 | 9.32   | +0.77        | deep_attn_2x8             | 5        | 13.99 | 8.92  | 0.01066 | 0.00923 | 12.66 | 9.32   | +3.34        |
| deep_2x8              | 6        | 8.85 | 8.92  | 0.01134 | 0.00923 | 10.29 | 9.34   | +0.95        | deep_attn_2x8             | 6        | 10.19 | 8.92  | 0.01014 | 0.00923 | 10.58 | 9.34   | +1.24        |
| deep_2x8              | 7        | 6.18 | 8.92  | 0.01076 | 0.00923 | 8.08  | 9.49   | <b>-1.41</b> | deep_attn_2x8             | 7        | 7.12  | 8.92  | 0.01016 | 0.00923 | 8.65  | 9.49   | <b>-0.83</b> |
| deep_2x8              | 12       | 7.03 | 8.92  | 0.01076 | 0.00923 | 8.78  | 9.56   | <b>-0.78</b> | deep_attn_2x8             | 12       | 7.57  | 8.92  | 0.01073 | 0.00923 | 9.17  | 9.56   | <b>-0.39</b> |
| deep_2x8              | 15       | 6.82 | 8.92  | 0.01050 | 0.00923 | 8.53  | 9.60   | <b>-1.07</b> | deep_attn_2x8             | 15       | 7.32  | 8.92  | 0.01028 | 0.00923 | 8.84  | 9.60   | <b>-0.76</b> |
| deep_2x8              | 18       | 6.74 | 8.92  | 0.01064 | 0.00923 | 8.51  | 9.62   | <b>-1.11</b> | deep_3x16 [dpyscf parity] | 7        | 7.44  | 8.92  | 0.00903 | 0.00923 | 8.46  | 9.49   | <b>-1.03</b> |
| deep_2x8              | 26       | 7.65 | 8.92  | 0.01015 | 0.00923 | 9.03  | 9.80   | <b>-0.77</b> | deep_3x16 [dpyscf parity] | 12       | 7.48  | 8.92  | 0.00892 | 0.00923 | 8.44  | 9.56   | <b>-1.12</b> |
| deep_3x16 [25 cycles] | 7        | 6.92 | 8.92  | 0.00972 | 0.00923 | 8.36  | 9.49   | <b>-1.13</b> | deep_3x16 [25 cycles]     | 12       | 7.17  | 8.92  | 0.00983 | 0.00923 | 8.57  | 9.56   | <b>-0.99</b> |

The same columns; **bold, negative** marks a cell below PBE: 15 of these 24 cells, 41 of the 64 over the two tables. deep\_2x8 is below PBE at every size but 3 to 6, where its energy leg is within 1.0 kcal/mol of PBE and its density leg 22 to 25 percent above; both arms are below PBE at 7 and 12.

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## Backup: the verdicts that depend on the leg or the units [holdout\_ed\_combined.csv]

| Choice                          | Cells whose verdict flips against the DFS-units, WTMAD-2 verdict              |
|---------------------------------|-------------------------------------------------------------------------------|
| plain-MAE energy leg            | deep_2x8 at 3, 4, 5 and 6; deep_attn_3x16 at 3, 5 and 6; deep0_attn_3x16 at 6 |
| self-calibrated grid-RMSE units | deep_2x8 at 3, 4, 5 and 6; deep_attn_3x16 at 2; deep0_3x16 at 3               |

Ten cells in all (deep\_2x8 at 3 to 6 flips under both choices); the other 54 of the 64 cells carry the same verdict under every choice, among them every deep\_3x16, deep\_attn\_2x8 and arm cell.

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